SAMPLE

Solar System Missions: Genesis

Collecting Solar Wind

Aira Astralis

Free sample: Introduction & Chapter 1
putrabooks.github.io

Introduction

In August 2001, a spacecraft named Genesis left Earth on a mission that sounds almost mythological: to fly closer to the Sun than any sample-collecting craft before it, catch the wind that blows from our star, and bring a piece of the Sun home. For more than two years it hovered a million and a half kilometers sunward of Earth, holding out collector arrays of the purest materials ever manufactured, and let the solar wind -- the perpetual stream of charged particles flowing off the Sun -- embed itself, ion by ion, into its wafers. Then it brought them back. The story of how those atoms were caught, how they survived a return to Earth that went spectacularly wrong, and what they told us about the birth of the solar system, is the subject of this book.

The premise of Genesis was simple and profound. The Sun contains 99.8 percent of the mass of the solar system. It is, in effect, the preserved sample of the cloud of gas and dust from which everything else -- planets, moons, asteroids, and us -- formed. We cannot go to the Sun and scoop up its matter, but the Sun comes to us: the solar wind carries atoms from the Sun's outer atmosphere outward past all the planets. Spacecraft had measured the solar wind's particles in flight for forty years, but in-flight instruments are small and their measurements coarse. Genesis was built to do the one thing no orbiting detector could: collect the wind and return it to laboratories on Earth, where the full arsenal of the world's most sensitive instruments could measure the atoms' chemical and isotopic fingerprints with a precision impossible in space.

Why does this matter? Because the Sun's composition is the starting composition of the solar system -- the baseline against which four and a half billion years of planetary chemistry must be measured. Planets are the Sun's raw material, processed: cooked, fractionated, oxidized, and recombined. When we find that a planet's atoms differ from the Sun's, we have found a chemical story -- a record of how the raw material became a world. Genesis delivered exactly such a discovery: the oxygen of the Sun, it turned out, is measurably different from the oxygen of the Earth, the Moon, and the meteorites -- a difference that points to subtle chemical sorting in the ancient solar nebula, and that helped settle a decades-old puzzle about the birth of the solar system.

The Genesis mission was also an engineering milestone: the first spacecraft to collect material beyond the Earth-Moon system and return it to Earth -- a feat the United States had last performed with Apollo 17 in 1972. It opened the modern era of robotic sample return, the era of Stardust, Hayabusa, Hayabusa2, and OSIRIS-REx, in which spacecraft routinely close the loop between exploration and laboratory science.

This book follows the standard structure of the series. We begin with discovery: the long road from the dream of sampling the Sun to the selection of Genesis in NASA's Discovery program and its launch in 2001. We explain the mission itself -- its spacecraft, its station at the Sun-Earth Lagrange point, its collector arrays, and the physics of catching solar wind. We walk through the mission's key characteristics, the science it was designed to do, the way its samples were observed and measured, and the crash of the sample capsule in the Utah desert in 2004 -- one of the most dramatic salvage operations in the history of space science. We place Genesis in context among the sample-return and heliophysics missions of its era, summarize the modern understanding it produced, count its significant discoveries, and face its challenges and remaining mysteries honestly. We close with future directions and the trivia shelf -- and with a Conclusion on what it means, in an age of far-flung exploration, that we have held atoms of the Sun in our hands.

Chapter 1: The Discovery of Genesis

The mission that became Genesis began, like most great missions, as an argument about atoms. By the 1970s, planetary scientists had a puzzle on their hands: the isotopes of oxygen. Isotopes are the chemically identical but differently weighted versions of an element -- oxygen-16, oxygen-17, and oxygen-18 -- and every natural process that makes or alters matter leaves its fingerprint in their relative abundances. Meteorites, the leftover rubble of the early solar system, showed a startling pattern: tiny grains inside them, the calcium-aluminum-rich inclusions, carried oxygen with a distinctly different isotopic mix than the Earth, the Moon, and most rocky material of the inner solar system. Either the ancient solar nebula was profoundly heterogeneous -- a mix of material from different stellar sources, never fully stirred -- or some chemical process had sorted the isotopes of oxygen after the Sun formed. Which was it? Nobody could say, because nobody knew the isotopic composition of the one object that had preserved the nebula unchanged: the Sun.

The problem was that knowing the Sun's isotopes was nearly impossible. Spectroscopy -- reading the Sun's light -- was powerful but could not deliver isotope ratios at the needed precision. In-situ spacecraft instruments, flown on

heliophysics missions through the solar wind since the 1960s, could count solar wind ions, but their resolution was orders of magnitude too coarse. What was needed was not a better instrument in space, but a sample: solar matter, physically present in an Earth laboratory. The solar wind, it was realized in the 1980s, made this possible. The wind is a direct sample of the Sun's outer atmosphere, accelerated outward and delivered, free of charge, to any spacecraft that cares to expose a clean surface to it. The idea took shape -- championed most persistently by the cosmochemist Donald Burnett of Caltech -- of a mission that would fly beyond the shielding of Earth's magnetic field, expose ultra-pure collector wafers to the solar wind for years, and bring them home for the world's mass spectrometers.

The intellectual heritage is worth pausing on. The Apollo program had already shown, in the 1960s and 1970s, what returned samples could do for lunar science -- a half-ton of Moon rock that planetary scientists still study today. Apollo astronauts had even collected solar wind on foil exposed at the surface of the Moon, and those foils, brought back for analysis, were the first solar wind samples ever measured in laboratories. But Apollo's foils were exposed only briefly, at one location, with materials of ordinary purity. What the new idea demanded was a purpose-built mission: months or years of collection, materials of extraordinary purity, and collectors designed for the highest-precision isotopic analysis.

The opportunity came with NASA's Discovery program -- the agency's "faster, better, cheaper" line of cost-capped, competitively selected science missions, created in the early 1990s. Burnett and colleagues proposed a solar-wind sample return mission, and in 1997 NASA selected it for flight, with Burnett as Principal Investigator. The mission's name, fitting for a spacecraft meant to recover the starting material of the solar system, was Genesis. JPL managed the project; the spacecraft and sample return capsule were built by Lockheed Martin Astronautics in Denver, and the mission's science team drew in cosmochemists from across the United States and Europe. Its total cost, by Discovery standards, was modest -- about a quarter of a billion dollars -- for a scientific payoff its proposers compared, without much exaggeration, to a piece of the Sun.

The mission profile set Genesis apart from everything else flying. To collect solar wind unaltered by Earth's magnetic field -- which would deflect the wind and contaminate a sample with terrestrial particles -- the spacecraft would travel not to a planet but to a place: the Sun-Earth L1 Lagrange point, the gravitational balance point about one and a half million kilometers sunward of Earth, where a spacecraft can hover with minimal fuel and a continuous, unobstructed view of the Sun. There it would expose its collectors, monitoring the wind's speed and character, for at least two years. Then it would fold its arrays into a sample return capsule, fire its engines, loop back -- and, in a first for robotic spacecraft, deliver extraterrestrial material collected in deep space to the desert of Utah, by parachute, for laboratory science.

On August 8, 2001, Genesis launched from Cape Canaveral on a Delta II rocket. Everything about the mission that followed -- the collection at L1, the return, the crash in Utah, the salvage, and the discoveries -- was still ahead. But the idea that had driven fifteen years of proposals had survived the journey from argument to orbit: the Sun could be sampled, and the sample brought home. What Genesis did with that opportunity -- and what its atoms taught us -- fills the rest of this book.

A note on the mission's place in the history of ideas belongs here, because Genesis was the closing move of a very old human ambition: to know what the Sun is made of. The ancients guessed fire and divinity; the spectroscope, in the nineteenth century, revealed the elements and founded astrophysics; the space age measured the wind in flight. Genesis completed the sequence: it made the Sun's substance tangible, subject to the same laboratory methods that analyze any rock on Earth. The mission's deepest rationale was that the Sun, our home star, should be treated as a sample -- and the discipline that treats stars as samples, cosmochemistry, gained from Genesis its anchor specimen. Every star studied by astronomy is now, in a small way, compared with the one star whose atoms we have held.

The Discovery program's competitive tradition also shaped the mission's spirit. Genesis was proposed by a scientist, not designed by a committee: its architecture was the simplest possible solution to its single question, and its science team was small enough to know every wafer by name. That intimacy had consequences the mission would come to appreciate: when the capsule broke, the people who cataloged its fragments were the same people who had polished its wafers before launch -- and their years of pre-flight documentation, born of affection as much as procedure, became the salvage's greatest asset.

The mission also closed a specific gap in the Apollo legacy. The Apollo solar wind foils -- gold and aluminum sheets exposed at the lunar surface and returned for analysis -- had proved the concept decades before Genesis flew, and their analysis had long been the best available solar wind sample measurement. Genesis's collectors

were, in a real sense, the answer to a question the Apollo foils had asked: what could a dedicated, purpose-built solar wind sample return achieve with years of exposure at L1, materials of far greater purity, and collectors designed with the laboratory's needs in mind? The two programs now stand in an unbroken line of solar wind sample science, one that began with astronauts and continued with a spacecraft that never came closer to the Sun than a million and a half kilometers -- and did not need to.